Tier 5 | Operate and recover

Identity and isolation

Authenticate a token, derive tenant and role on the server, then AND tenant scope with user filters. Deny unauthorized mutations before database access. Include tenant, model space and index generation in cache identity.

Durability is a measured boundary

Log before acknowledging writes. Checkpoint with a log watermark and replay after restart. Test abrupt exit and torn tails. A successful process-crash test does not establish power-loss or distributed guarantees.

Migrate embeddings safely

Create a new space/collection, backfill, replay changes, validate quality and deletion state, switch an alias, then retain a rollback window. Never overwrite an old space merely because the new model has the same dimension.

Backup and deletion

Define exactly which data a snapshot covers. Keep a deletion ledger and replay it during restore. Measure recovery time and acknowledged writes lost; the course snapshot drill deliberately demonstrates a five-write gap.

Load and monitoring

Use bounded concurrency, record errors and p95/p99, and distinguish serialized QPS from concurrent throughput. Observe queueing and resource saturation. Inject a process failure and measure recovery rather than hiding failed requests.

Run and clean up

uv run --extra ops python scripts/prepare_tier5.py
Follow docs/TIER5_RUNBOOK.md for TLS startup and credentials. Run scripts/cleanup.py to preview owned resources before removal. Do not print tokens or use global container prune.